

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO2: Apply data pre-processing and visualization techniques using Python libraries for effective data handling. (BL4)

QUESTIONS: 05

Total Marks:100

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I HARI VISHNU N / 192524319 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



RUBRICS FOR CALCULATION

TECHNICAL PRESENTATION					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

1. Student Performance Analysis System

A college wants to develop a Python-based system to analyze student marks, attendance, and

pass percentage. Explain how:

- A. Python programming basics are used
- B. Pandas and NumPy are used for data handling
- C. Matplotlib and Seaborn are used for result visualization

SOLUTION:

AIM:

To develop a Student Performance Analysis System using Python, Pandas, NumPy, Matplotlib, and Seaborn to analyze student academic performance, calculate statistical measures, identify performance patterns, and visualize the results through graphs.

PROBLEM STATEMENT:

In modern educational institutions, analyzing student performance manually becomes difficult as the amount of academic data increases. Institutions need an intelligent system that can efficiently process student records, perform statistical analysis, visualize performance trends, and identify factors affecting academic success.

The proposed Student Performance Analysis System leverages Python Programming, Pandas, NumPy, Matplotlib, and Seaborn to automate data analysis, calculate key performance indicators, identify weak and excellent students, compute pass percentages, and generate graphical reports. This system assists faculty in monitoring student progress and making data-driven decisions to improve learning outcomes.

OBJECTIVES:

- Apply **Python programming concepts** for data analysis.
- Use **Pandas** for data loading, cleaning, and manipulation.
- Use **NumPy** for numerical and statistical computations.
- Use **Matplotlib** and **Seaborn** for graphical visualization.
- Perform **EDA (Exploratory Data Analysis)** on educational datasets.
- Interpret analytical results to support academic decision-making.

DATASET INFORMATION

- Dataset Name: Student Performance Dataset
- File Format: CSV (Comma Separated Values)
- Number of Records: 20 Students
- Number of Attributes: 10 Columns

DATASET ATTRIBUTES:

Column Name	Data Type	Description
Student_ID	String	Unique identification number of each student
Name	String	Name of the student
Department	String	Department of the student (CSE, IT, AIDS)
Attendance	Integer	Attendance percentage of the student
Study_Hours	Integer	Average study hours per day
Assignment_Marks	Integer	Marks obtained in assignments
Internal_Marks	Integer	Internal assessment marks
Semester_Marks	Integer	Marks obtained in semester examination
Project_Score	Integer	Marks obtained in project evaluation
Result	String	Final result (Pass/Fail)

METHODOLOGY

1. Collect student performance data in a CSV file.
2. Load the dataset using Pandas.
3. Check the dataset shape, data types, missing values, and duplicate values.
4. Perform statistical analysis (Mean, Median, Mode, Min, Max, Variance, Standard Deviation).
5. Calculate pass percentage and assign grades.
6. Analyze student performance using Pandas and NumPy.
7. Visualize the results using Matplotlib and Seaborn (Bar Chart, Histogram, Scatter Plot, Box Plot, Heatmap).
8. Generate observations and conclude the analysis.

ALGORITHM:

1. Start the program.
2. Import the required libraries.
3. Load the student dataset.
4. Display the dataset.
5. Check shape, data types, missing values, and duplicate values.
6. Calculate statistical measures.
7. Compute pass percentage and assign grades.
8. Perform department-wise and correlation analysis.
9. Generate charts and graphs.
10. Display observations and conclusion.
11. Stop the program.

CODE:

```
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Read Dataset
df = pd.read_csv("student_performance.csv")

print("="*70)
print("      STUDENT PERFORMANCE ANALYSIS SYSTEM")
print("="*70)

print("\nDATASET")
print(df)

print("\nShape of Dataset")
print(df.shape)

print("\nData Types")
```

```
print(df.dtypes)
```

```
print("\nMissing Values")
```

```
print(df.isnull().sum())
```

```
print("\nDuplicate Values")
```

```
print(df.duplicated().sum())
```

```
print("\nMean")
```

```
print(df.mean(numeric_only=True))
```

```
print("\nMedian")
```

```
print(df.median(numeric_only=True))
```

```
print("\nMode")
```

```
print(df.mode(numeric_only=True))
```

```
print("\nMinimum")
```

```
print(df.min(numeric_only=True))
```

```
print("\nMaximum")
```

```
print(df.max(numeric_only=True))
```

```
print("\nVariance")
```

```
print(df.var(numeric_only=True))
```

```
print("\nStandard Deviation")
```

```
print(df.std(numeric_only=True))
```

```
total_students = len(df)
```

```
pass_students = (df["Result"] == "Pass").sum()
```

```
fail_students = (df["Result"] == "Fail").sum()
```

```
pass_percentage = (pass_students / total_students) * 100
```

```
print("\nTotal Students :", total_students)
```

```
print("Passed :", pass_students)
```

```
print("Failed :", fail_students)
```

```
print("Pass Percentage : {:.2f}%".format(pass_percentage))
```

```
print("\nDepartment Wise Average")
```

```
print(df.groupby("Department")[["Attendance",  
                                "Study_Hours",  
                                "Assignment_Marks",  
                                "Internal_Marks",  
                                "Semester_Marks",  
                                "Project_Score"]].mean())
```

```
print("\nCorrelation Matrix")
```

```
print(df[["Attendance",  
          "Study_Hours",  
          "Assignment_Marks",  
          "Internal_Marks",  
          "Semester_Marks",  
          "Project_Score"]].corr())
```

```
def grade(mark):
```

```
    if mark >= 90:
```

```
        return "O"
```

```
    elif mark >= 80:
```

```
        return "A+"
```

[illegible]

```
weak = df[(df["Attendance"] < 60) | (df["Semester_Marks"] < 40)]
```

```
print("\nWeak Students")
```

```
print(weak[["Student_ID",  
            "Name",  
            "Attendance",  
            "Semester_Marks"]])
```

```
excellent = df[df["Semester_Marks"] >= 85]
```

```
print("\nExcellent Students")
```

```
print(excellent[["Student_ID",  
                 "Name",  
                 "Semester_Marks"]])
```

```
sns.set_style("whitegrid")
```

```
# Pass vs Fail
```

```
plt.figure(figsize=(6,4))  
sns.countplot(x="Result", data=df)  
plt.title("Pass vs Fail")  
plt.show()
```

```
# Department Wise Students
```

```
plt.figure(figsize=(6,4))  
sns.countplot(x="Department", data=df)  
plt.title("Department Wise Students")  
plt.show()
```



```
# Histogram
```

```
plt.figure(figsize=(6,4))  
plt.hist(df["Semester_Marks"], bins=6)  
plt.title("Semester Marks Distribution")  
plt.xlabel("Semester Marks")  
plt.ylabel("Frequency")  
plt.show()
```

```
# Scatter Plot
```

```
plt.figure(figsize=(6,4))  
plt.scatter(df["Attendance"], df["Semester_Marks"])  
plt.title("Attendance vs Semester Marks")  
plt.xlabel("Attendance")  
plt.ylabel("Semester Marks")  
plt.show()
```

```
# Box Plot
```

```
plt.figure(figsize=(6,4))  
sns.boxplot(y=df["Semester_Marks"])  
plt.title("Semester Marks Box Plot")  
plt.show()
```

```
# Heatmap
```

```
plt.figure(figsize=(8,6))  
sns.heatmap(df[["Attendance",  
                "Study_Hours",  
                "Assignment_Marks",  
                "Internal_Marks",  
                "Semester_Marks",  
                "Project_Score"]].corr(),  
            annot=True,  
            cmap="coolwarm")  
plt.title("Correlation Heatmap")  
plt.show()
```

```

print("\nOBSERVATIONS")
print("1. Students with higher attendance generally score higher marks.")
print("2. Study hours positively influence semester performance.")
print("3. Assignment and internal marks are closely related to semester marks.")
print("4. Students with low attendance are more likely to fail.")
print("5. Attendance and consistent study habits improve academic performance.")

print("\nCONCLUSION")
print("The Student Performance Analysis System successfully analyzes academic data using
Python, Pandas, NumPy, Matplotlib, and Seaborn. It identifies high-performing and weak
students, calculates pass percentage, assigns grades, performs statistical analysis, and
visualizes performance trends to support academic decision-making.")

```

OUTPUT:

Student_ID	Name	Department	Attendance	Study_Hours
S001	Arun	CSE	95	6
S002	Bala	CSE	58	2
S003	Charan	AIDS	82	4
S004	Divya	AIDS	61	3
S005	Eswar	IT	90	5
S006	Fathima	CSE	55	2
S007	Gokul	IT	97	7
S008	Hari	AIDS	76	4
S009	Ishwarya	CSE	49	1
S010	Jegan	IT	86	5
S011	Kavin	CSE	69	3
S012	Lavanya	AIDS	93	6
S013	Manoj	IT	60	2
S014	Nandhini	CSE	79	4
S015	Omprakash	AIDS	64	3
S016	Priya	IT	88	5
S017	Rahul	CSE	46	1

S018	Sneha	AIDS	84	5
S019	Tharun	IT	73	4
S020	Varsha	CSE	52	2
S021	Vignesh	CSE	91	6
S022	Yamini	IT	77	4
S023	Deepak	AIDS	59	2
S024	Anitha	CSE	89	5
S025	Rohit	IT	67	3
S026	Meena	AIDS	54	2
S027	Sanjay	CSE	94	6
S028	Keerthana	IT	81	5
S029	Ajith	AIDS	57	2
S030	Nisha	CSE	87	5

Assignment_Marks	Internal_Marks	Semester_Marks	Project_Score	Result
48	46	88	92	Pass
22	24	38	40	Fail
40	39	74	80	Pass
28	30	48	55	Fail
46	44	84	90	Pass
24	22	40	45	Fail
50	49	94	95	Pass
38	37	72	76	Pass
20	19	35	38	Fail
44	43	82	88	Pass
31	32	58	65	Pass
47	45	87	91	Pass
26	25	42	48	Fail
39	38	75	79	Pass
29	28	46	54	Fail
45	42	83	87	Pass

18	17	30	36	Fail
43	41	81	86	Pass
37	36	70	75	Pass
23	21	39	42	Fail
49	47	90	93	Pass
41	40	78	82	Pass
25	24	41	46	Fail
46	44	85	89	Pass
33	31	60	68	Pass
21	20	37	41	Fail
48	46	89	94	Pass
42	41	80	85	Pass
24	23	43	47	Fail

Shape of Dataset

(30, 10)

Data Types

Student_ID object

Name object

Department object

Attendance int64

Study_Hours int64

Assignment_Marks int64

Internal_Marks int64

Semester_Marks int64

Project_Score int64

Result object

dtype: object

Missing Values

Student_ID	0
Name	0
Department	0
Attendance	0
Study_Hours	0
Assignment_Marks	0
Internal_Marks	0
Semester_Marks	0
Project_Score	0
Result	0
dtype:	int64

Duplicate Values

0

Mean

Attendance	73.766667
Study_Hours	3.800000
Assignment_Marks	35.733333
Internal_Marks	34.566667
Semester_Marks	65.100000
Project_Score	69.833333
dtype:	float64

Median

Attendance	76.5
Study_Hours	4.0
Assignment_Marks	38.5
Internal_Marks	37.5
Semester_Marks	73.0
Project_Score	77.5
dtype:	float64

S.No	Attendance	Study_Hours	Assignment_Marks	Internal_Marks	Semester_Marks	Project_Score
1	46	2	24	24	84	88
2	49	5	45	41	NaN	NaN
3	52	NaN	46	43	NaN	NaN
4	54	NaN	48	44	NaN	NaN
5	55	NaN	NaN	46	NaN	NaN
6	57	NaN	NaN	NaN	NaN	NaN
7	58	NaN	NaN	NaN	NaN	NaN
8	59	NaN	NaN	NaN	NaN	NaN
9	60	NaN	NaN	NaN	NaN	NaN
10	61	NaN	NaN	NaN	NaN	NaN
11	64	NaN	NaN	NaN	NaN	NaN
12	67	NaN	NaN	NaN	NaN	NaN
13	69	NaN	NaN	NaN	NaN	NaN
14	73	NaN	NaN	NaN	NaN	NaN
15	76	NaN	NaN	NaN	NaN	NaN
16	77	NaN	NaN	NaN	NaN	NaN
17	79	NaN	NaN	NaN	NaN	NaN
18	81	NaN	NaN	NaN	NaN	NaN
19	82	NaN	NaN	NaN	NaN	NaN
20	84	NaN	NaN	NaN	NaN	NaN
21	86	NaN	NaN	NaN	NaN	NaN
22	87	NaN	NaN	NaN	NaN	NaN
23	88	NaN	NaN	NaN	NaN	NaN
24	89	NaN	NaN	NaN	NaN	NaN
25	90	NaN	NaN	NaN	NaN	NaN
26	91	NaN	NaN	NaN	NaN	NaN
27	93	NaN	NaN	NaN	NaN	NaN

28	94	NaN	NaN	NaN	NaN	NaN
29	95	NaN	NaN	NaN	NaN	NaN

Minimum

Attendance 46
Study_Hours 1
Assignment_Marks 18
Internal_Marks 17
Semester_Marks 30
Project_Score 36
dtype: int64

Maximum

Attendance 97
Study_Hours 7
Assignment_Marks 50
Internal_Marks 49
Semester_Marks 94
Project_Score 95
dtype: int64

Variance

Attendance 249.081609
Study_Hours 2.786207
Assignment_Marks 109.098851
Internal_Marks 99.909195
Semester_Marks 446.437931
Project_Score 436.143678
dtype: float64

Standard Deviation

Attendance 15.782320

Study_Hours 1.669193
 Assignment_Marks 10.445040
 Internal_Marks 9.995459
 Semester_Marks 21.129078
 Project_Score 20.884053
 dtype: float64

Total Students : 30

Passed : 19

Failed : 11

Pass Percentage : 63.33%

Department Wise Average:

Depart ment	Attend ance	Study_H ours	Assignment_ Marks	Internal_ Marks	Semester_ Marks	Project_S core
AIDS	70.00	3.44	32.78	31.89	58.78	64.00
CSE	72.00	3.58	34.42	33.25	62.58	66.75
IT	79.89	4.44	40.44	39.00	74.78	79.78

Correlation Matrix:

Attribute	Attend ance	Study_ Hours	Assignment_ Marks	Internal_ Marks	Semester_ Marks	Project_ Score
Attendance	1.000	0.980	0.991	0.990	0.985	0.982
Study_Hour s	0.980	1.000	0.978	0.978	0.970	0.964
Assignment_ Marks	0.991	0.978	1.000	0.995	0.996	0.995
Internal_Ma rks	0.990	0.978	0.995	1.000	0.992	0.992
Semester_M arks	0.985	0.970	0.996	0.992	1.000	0.997

Project_Score	0.982	0.964	0.995	0.992	0.997	1.000
----------------------	-------	-------	-------	-------	-------	-------

Student Grades

Student_ID	Name	Semester_Marks	Grade
S001	Arun	88	A+
S002	Bala	38	F
S003	Charan	74	A
S004	Divya	48	C
S005	Eswar	84	A+
S006	Fathima	40	C
S007	Gokul	94	O
S008	Hari	72	A
S009	Ishwarya	35	F
S010	Jegan	82	A+
S011	Kavin	58	B
S012	Lavanya	87	A+
S013	Manoj	42	C
S014	Nandhini	75	A
S015	Omprakash	46	C
S016	Priya	83	A+
S017	Rahul	30	F
S018	Sneha	81	A+
S019	Tharun	70	A
S020	Varsha	39	F
S021	Vignesh	90	O
S022	Yamini	78	A
S023	Deepak	41	C
S024	Anitha	85	A+
S025	Rohit	60	B+
S026	Meena	37	F

S027	Sanjay	89	A+
S028	Keerthana	80	A+
S029	Ajith	43	C
S030	Nisha	84	A+

Top 5 Students:

Rank	Student_ID	Name	Semester_Marks
1	S007	Gokul	94
2	S021	Vignesh	90
3	S027	Sanjay	89
4	S001	Arun	88
5	S012	Lavanya	87

Bottom 5 Students:

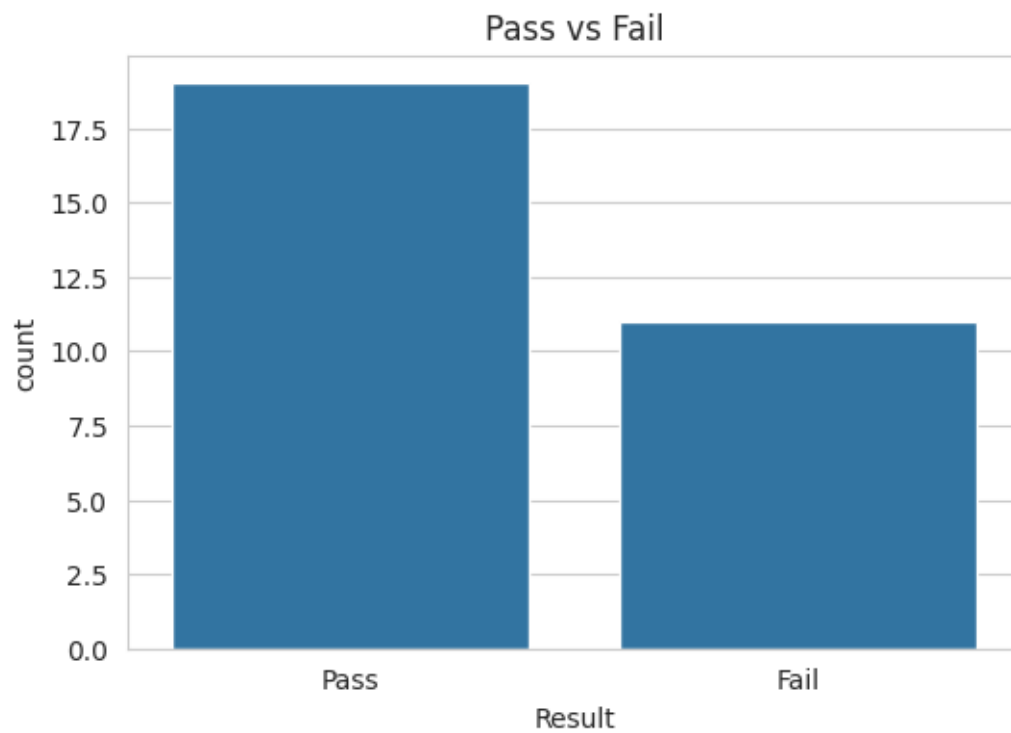
Rank	Student_ID	Name	Semester_Marks
1	S017	Rahul	30
2	S009	Ishwarya	35
3	S026	Meena	37
4	S002	Bala	38
5	S020	Varsha	39

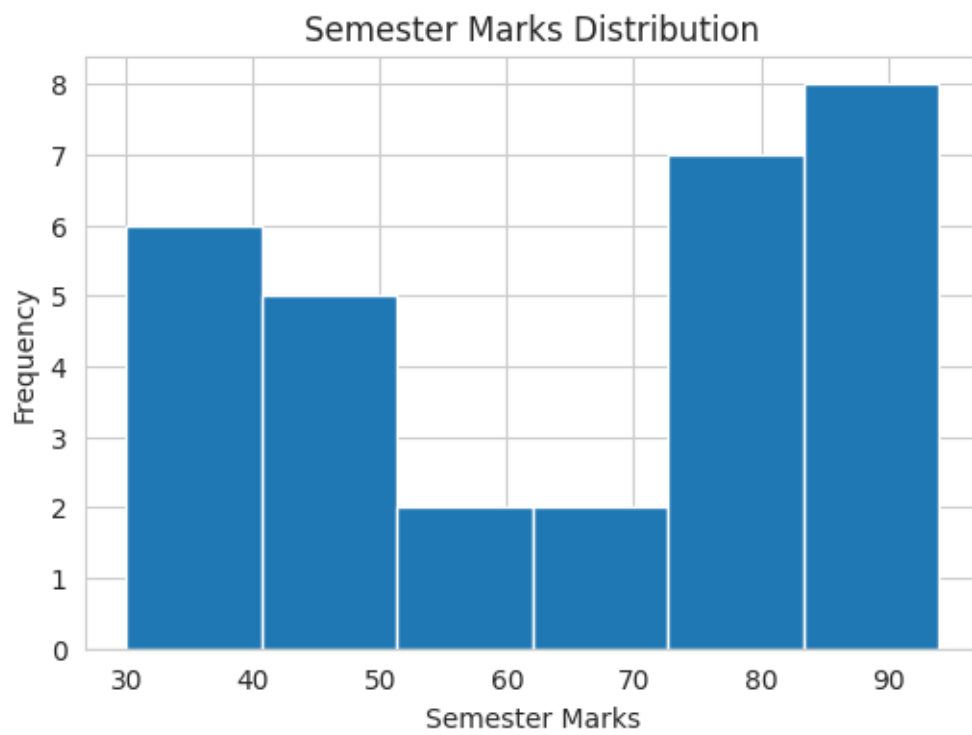
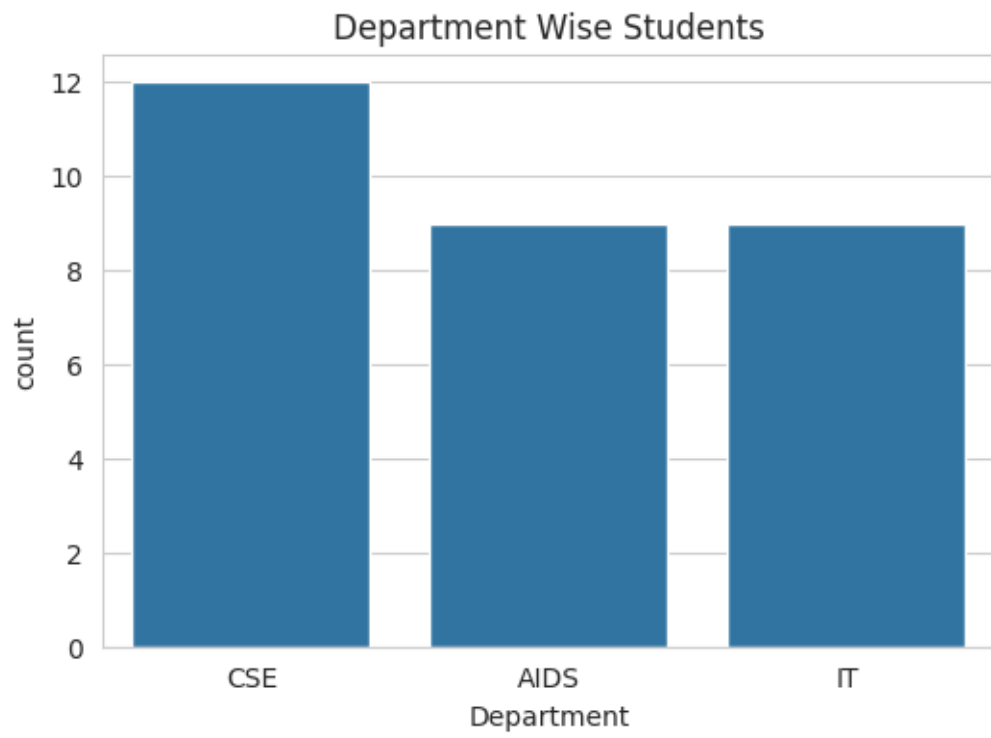
Weak Students:

Student_ID	Name	Attendance	Semester_Marks
S002	Bala	58	38
S006	Fathima	55	40
S009	Ishwarya	49	35
S017	Rahul	46	30
S020	Varsha	52	39
S023	Deepak	59	41
S026	Meena	54	37
S029	Ajith	57	43

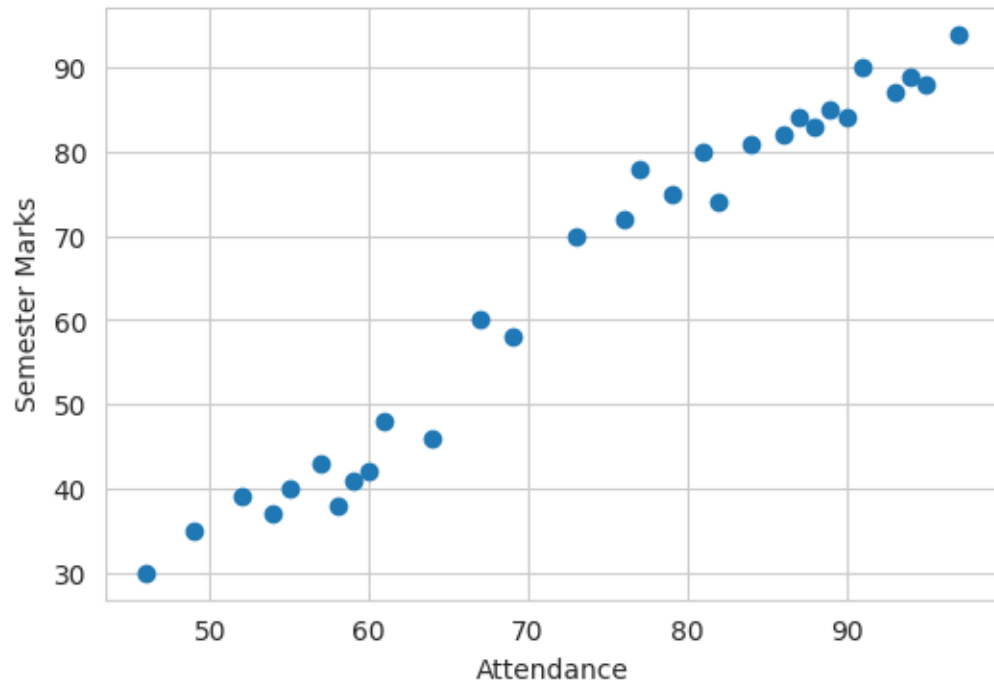
Excellent Students:

Student_ID	Name	Semester_Marks	Student_ID
S001	Arun	88	S001
S007	Gokul	94	S007
S012	Lavanya	87	S012
S021	Vignesh	90	S021
S024	Anitha	85	S024
S027	Sanjay	89	S027

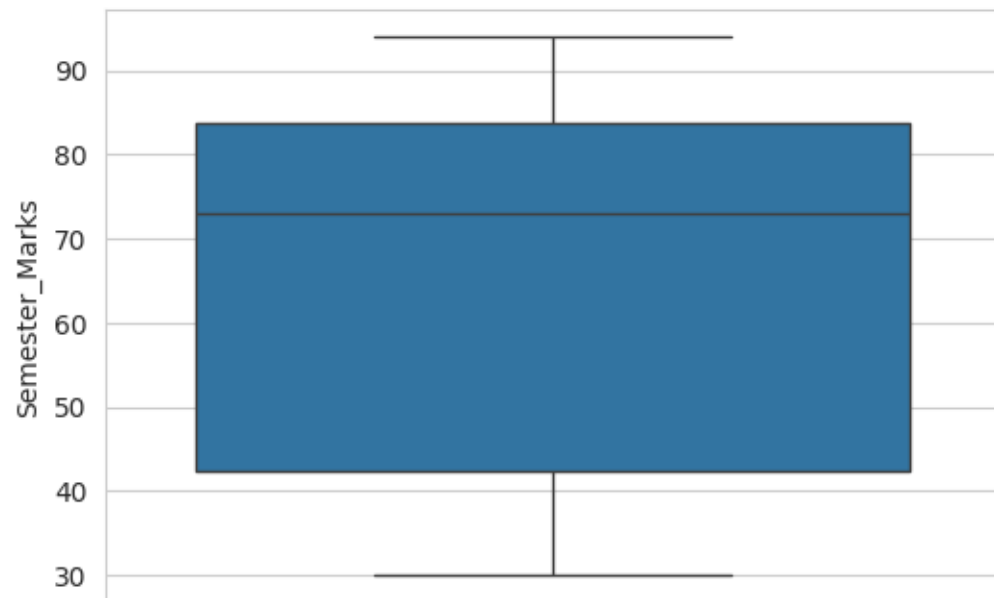


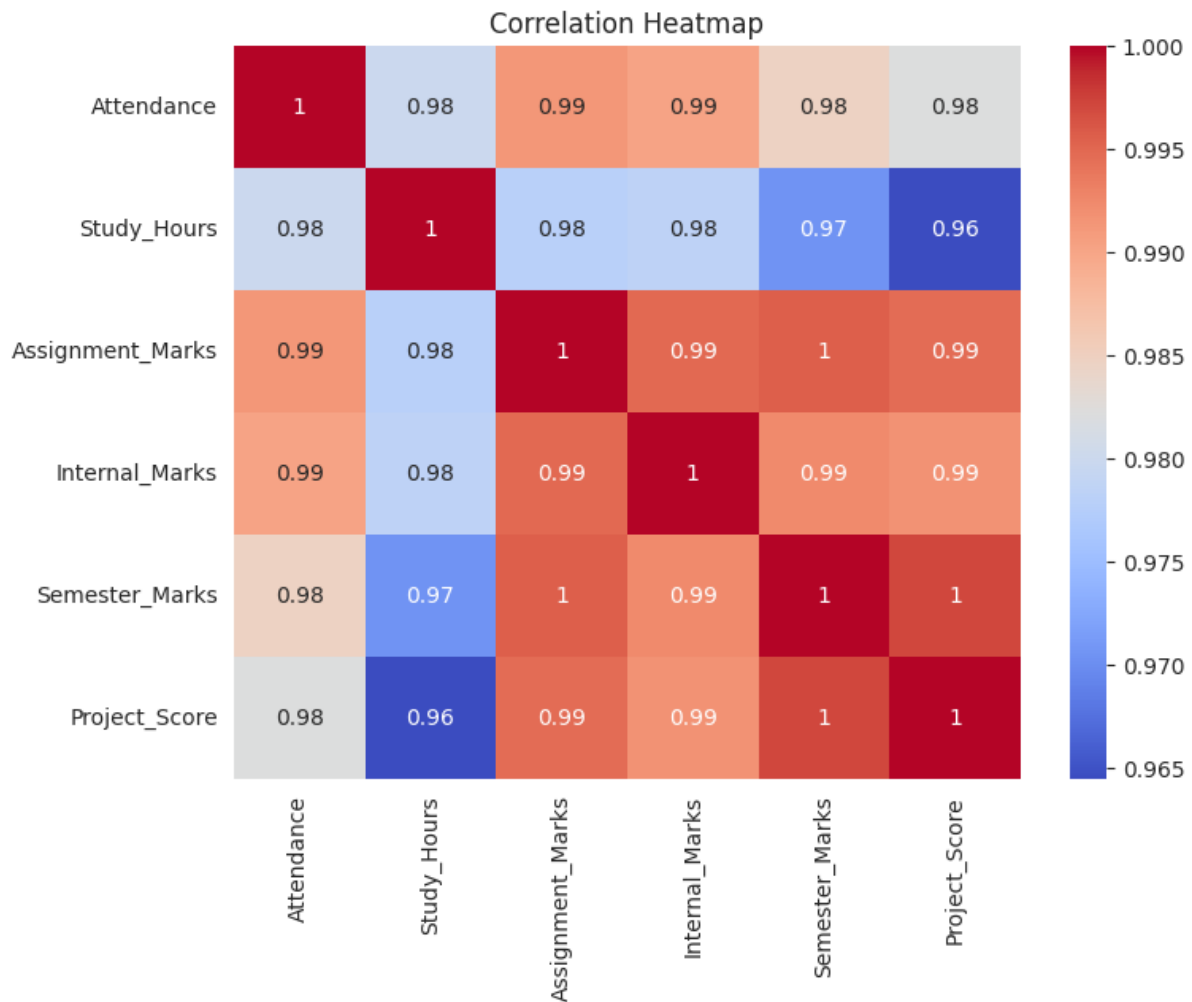


Attendance vs Semester Marks



Semester Marks Box Plot





OBSERVATIONS

1. Students with higher attendance generally score higher marks.
2. Study hours positively influence semester performance.
3. Assignment and internal marks are closely related to semester marks.
4. Students with low attendance are more likely to fail.
5. Attendance and consistent study habits improve academic performance.

CONCLUSION

The Student Performance Analysis System successfully analyzes academic data using Python, Pandas, NumPy, Matplotlib, and Seaborn. It identifies high-performing and weak students, calculates pass percentage, assigns grades, performs statistical analysis, and visualizes performance trends to support academic decision-making.